

Practical Assignment: Regression Analysis

2026-06-24

Date: 26 June 2026

Time: 10:30 a.m. – 3:15 p.m.

(The practical session consists of two sessions with a lunch break in between.

To make effective use of the practical session, students are encouraged to begin data exploration, preprocessing and preliminary analysis of the **training dataset** before attending the session on 26 June 2026.

About Data

This practical activity is based on data related to the **Islandwide Water Level and Rainfall Situation in Major Rivers of Sri Lanka**.

The training dataset contains the following variables:

Variable	Description
Gauging_station	Identifier of the gauging station
Water_Level_Xt_1	River water level (mm) at time t
24HrRF_Xt_1	Total rainfall (mm) received during the previous 24 hours at time t
Water_Level_Xt	River water level (mm) at time t + 12 hours (target variable)

- The objective is to develop a regression model capable of predicting the river water level 12 hours ahead.
- The training dataset has been provided and should be used for model development and evaluation.
- A separate **test dataset** will be uploaded on 26 June 2026. This dataset will not contain the target variable and should be used to generate predictions using your final model.

Task 1

Conduct a complete regression analysis (Both univariate and multivariate) using the training dataset.

Your analysis should include the following:

1. Exploratory data analysis
2. Model development
3. Model diagnostics
4. Model evaluation
5. Discuss the practical usefulness of the model for river water level forecasting.

6. Discuss strengths and limitations of the model.

Prepare your analysis in a Jupyter Notebook.

Submit:

1. The Jupyter Notebook (.ipynb) file.
2. The HTML document generated from the notebook.

The notebook should be clearly organized, well documented and include explanations of all major steps and findings.

Task 2: Forecast generation

A test dataset will be released on 26 June 2026.

Using the final model developed in Task 1:

1. Generate predictions for `Water_Level_Xt` for all observations in the test dataset.
2. Save the predictions in the format specified during the practical session.
3. Submit the prediction file before the deadline announced during the session.